

1st Part Problem Set for Applied Abstract Algebra

Quivers: Representations and Morphisms

1. Let $M, M' \in \text{Rep } Q$. Show that the set of morphisms $\text{Hom}(M, M')$ is a k -vector space.

Proof. Let $M = (M_i, \phi_\alpha)$ and $M' = (M'_i, \psi_\alpha)$. $\forall f, g \in \text{Hom}(M, M')$ since f, g are morphisms, for any arrow $\alpha : i \rightarrow j$, we have $\psi_\alpha f_i = f_j \phi_\alpha$ and $\psi_\alpha g_i = g_j \phi_\alpha$.

(a) **Addition:**

$$\psi_\alpha(f + g)_i = \psi_\alpha(f_i + g_i) = \psi_\alpha f_i + \psi_\alpha g_i = f_j \phi_\alpha + g_j \phi_\alpha = (f_j + g_j) \phi_\alpha = (f + g)_j \phi_\alpha.$$

Thus $f + g \in \text{Hom}(M, M')$.

(b) **Scalar Multiplication:**

$$\psi_\alpha(\lambda f)_i = \psi_\alpha(\lambda f_i) = \lambda(\psi_\alpha f_i) = \lambda(f_j \phi_\alpha) = (\lambda f_j) \phi_\alpha = (\lambda f)_j \phi_\alpha.$$

Thus $\lambda f \in \text{Hom}(M, M')$.

The zero morphism $0 = (0)_{i \in Q_0}$ clearly satisfies the commutativity condition. The vector space axioms follow from the structure of linear maps between vector spaces. $\square$

2. Let Q be the quiver

$$1 \longrightarrow 2 \longleftarrow 3$$

and consider the representations

$$M: \quad k \xrightarrow{\begin{pmatrix} 1 \\ 0 \end{pmatrix}} k^2 \xleftarrow{\begin{pmatrix} 0 \\ 1 \end{pmatrix}} k \quad \quad M': \quad k \xrightarrow{\begin{pmatrix} 1 \\ 0 \end{pmatrix}} k^2 \xleftarrow{\begin{pmatrix} 1 \\ 0 \end{pmatrix}} k$$

(a) Show that M and M' are not indecomposable.

Proof. We exhibit a nontrivial direct sum decomposition for each representation.

Decomposition of M . Define subrepresentations

$$N_1: \quad k \xrightarrow{\begin{pmatrix} 1 \\ 0 \end{pmatrix}} \text{span}(e_1) \xleftarrow{0} 0, \quad \quad N_2: \quad 0 \xrightarrow{0} \text{span}(e_2) \xleftarrow{\begin{pmatrix} 0 \\ 1 \end{pmatrix}} k.$$

At each vertex we have $M(1) = k = N_1(1) \oplus N_2(1)$, $M(2) = k^2 = \text{span}(e_1) \oplus \text{span}(e_2) = N_1(2) \oplus N_2(2)$, and $M(3) = k = N_1(3) \oplus N_2(3)$. The linear maps of M respect this decomposition: the map $\begin{pmatrix} 1 \\ 0 \end{pmatrix}: k \rightarrow k^2$ lands in $\text{span}(e_1) = N_1(2)$, and

$\begin{pmatrix} 0 \\ 1 \end{pmatrix}: k \rightarrow k^2$ lands in $\text{span}(e_2) = N_2(2)$. Hence $M = N_1 \oplus N_2 \cong P(1) \oplus P(3)$, so M is not indecomposable.

Decomposition of M' . Define subrepresentations

$$N'_1: \quad k \xrightarrow{1} \text{span}(e_1) \xleftarrow{1} k, \quad N'_2: \quad 0 \xrightarrow{0} \text{span}(e_2) \xleftarrow{0} 0.$$

At each vertex: $M'(1) = k = N'_1(1) \oplus N'_2(1)$, $M'(2) = k^2 = \text{span}(e_1) \oplus \text{span}(e_2) = N'_1(2) \oplus N'_2(2)$, and $M'(3) = k = N'_1(3) \oplus N'_2(3)$. Both arrows $\begin{pmatrix} 1 \\ 0 \end{pmatrix}: M'(1) \rightarrow M'(2)$ and $\begin{pmatrix} 1 \\ 0 \end{pmatrix}: M'(3) \rightarrow M'(2)$ land in $\text{span}(e_1) = N'_1(2)$, so the maps respect the decomposition. Hence $M' = N'_1 \oplus N'_2 \cong I(2) \oplus S(2)$, so M' is not indecomposable. $\square$

(b) Show that M and M' are not isomorphic.

Proof. Suppose for contradiction that $f: M \rightarrow M'$ is an isomorphism, so $f = (f_1, f_2, f_3)$ with f_i invertible. The commutativity conditions give:

$$f_2\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}f_1, \quad f_2\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}f_3.$$

Let $f_1 = (a) \in k^*$, $f_3 = (c) \in k^*$, and $f_2 = \begin{pmatrix} p & q \\ r & s \end{pmatrix} \in \text{GL}_2(k)$.

From the first equation: $f_2\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} p \\ r \end{pmatrix} = \begin{pmatrix} a \\ 0 \end{pmatrix}$, so $p = a$ and $r = 0$.

From the second equation: $f_2\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} q \\ s \end{pmatrix} = \begin{pmatrix} c \\ 0 \end{pmatrix}$, so $q = c$ and $s = 0$.

Then $f_2 = \begin{pmatrix} a & c \\ 0 & 0 \end{pmatrix}$, which has $\det(f_2) = 0$. This contradicts $f_2 \in \text{GL}_2(k)$.

Therefore $M \not\cong M'$. $\square$

3. Let Q be the quiver

$$1 \longrightarrow 2 \longleftarrow 3$$

and let M be the representation

$$M: \quad k^2 \xrightarrow{\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}} k^3 \xleftarrow{\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}} k.$$

(a) Write M as a direct sum of the indecomposable representations listed below:

$$\begin{array}{lll} S(1): k \rightarrow 0 \leftarrow 0 & S(2): 0 \rightarrow k \leftarrow 0 & S(3): 0 \rightarrow 0 \leftarrow k \\ P(1): k \xrightarrow{1} k \leftarrow 0 & I(2): k \xrightarrow{1} k \xleftarrow{1} k & P(3): 0 \rightarrow k \xleftarrow{1} k \end{array}$$

Proof. We claim that

$$M \cong I(2) \oplus P(1) \oplus S(2).$$

The dimension vectors of the summands are

$$\underline{\dim} I(2) = (1, 1, 1), \quad \underline{\dim} P(1) = (1, 1, 0), \quad \underline{\dim} S(2) = (0, 1, 0),$$

and their sum is

$$(1, 1, 1) + (1, 1, 0) + (0, 1, 0) = (2, 3, 1) = \underline{\dim} M.$$

We now verify that the maps of $I(2) \oplus P(1) \oplus S(2)$ agree with those of M . Order the components at each vertex as:

$$\text{Vertex 1: } I(2)(1) \oplus P(1)(1) = k \oplus k = k^2,$$

$$\text{Vertex 2: } I(2)(2) \oplus P(1)(2) \oplus S(2)(2) = k \oplus k \oplus k = k^3,$$

$$\text{Vertex 3: } I(2)(3) = k.$$

Map at arrow $1 \rightarrow 2$. In the direct sum, the map $k^2 \rightarrow k^3$ sends

$$(a, b) \mapsto (I(2)(\alpha)(a), P(1)(\alpha)(b), 0) = (a, b, 0),$$

since both $I(2)$ and $P(1)$ have identity maps at this arrow, and $S(2)(1) = 0$ contributes nothing. In matrix form this gives

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix},$$

which agrees with the map of M .

Map at arrow $3 \rightarrow 2$. In the direct sum, the map $k \rightarrow k^3$ sends

$$c \mapsto (I(2)(\beta)(c), 0, 0) = (c, 0, 0),$$

since $I(2)$ has identity map at this arrow, and $P(1)(3) = S(2)(3) = 0$ contribute nothing. In matrix form this gives

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix},$$

which agrees with the map of M . Therefore $M \cong I(2) \oplus P(1) \oplus S(2)$. $\square$

(b) Show that there is a non-split short exact sequence

$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0.$$

Proof. Consider the representations

$$S(2) = (0, k, 0), \quad P(1) = (k, k, 0) \text{ with map } \text{id}_k, \quad S(1) = (k, 0, 0).$$

Define morphisms $\iota: S(2) \rightarrow P(1)$ and $p: P(1) \rightarrow S(1)$ by

$$\iota = (\iota_1, \iota_2, \iota_3) = (0, \text{id}_k, 0), \quad p = (p_1, p_2, p_3) = (\text{id}_k, 0, 0).$$

Morphism verification. The only nontrivial arrow is $\alpha: 1 \rightarrow 2$ with $P(1)(\alpha) = \text{id}_k$. For ι , the commutativity diagram at α is

$$\begin{array}{ccc} S(2)(1) = 0 & \xrightarrow{0} & S(2)(2) = k \\ \iota_1=0 \downarrow & & \downarrow \iota_2=\text{id}_k \\ P(1)(1) = k & \xrightarrow{\text{id}_k} & P(1)(2) = k \end{array}$$

Both compositions give 0, so the diagram commutes. For p :

$$\begin{array}{ccc} P(1)(1) = k & \xrightarrow{\text{id}_k} & P(1)(2) = k \\ p_1=\text{id}_k \downarrow & & \downarrow p_2=0 \\ S(1)(1) = k & \xrightarrow{0} & S(1)(2) = 0 \end{array}$$

Both compositions give 0, so the diagram commutes.

Exactness.

ι is injective: the only nontrivial component is $\iota_2 = \text{id}_k$, which is injective.

p is surjective: $p_1 = \text{id}_k$ is surjective, and $p_2: k \rightarrow 0$, $p_3: 0 \rightarrow 0$ are trivially surjective.

Exactness at $P(1)$: we verify $\ker p = \text{im } \iota$ at each vertex.

- Vertex 1: $\ker(\text{id}_k) = 0 = \text{im}(0)$.
- Vertex 2: $\ker(0: k \rightarrow 0) = k = \text{im}(\text{id}_k)$.
- Vertex 3: both are 0.

Hence

$$0 \longrightarrow S(2) \xrightarrow{\iota} P(1) \xrightarrow{p} S(1) \longrightarrow 0$$

is a short exact sequence.

Non-split. If the sequence splits, then

$$P(1) \cong S(1) \oplus S(2).$$

However, the direct sum $S(1) \oplus S(2) = (k, k, 0)$ has map $0 \oplus 0 = 0$ at arrow $1 \rightarrow 2$, while $P(1)$ has map $\text{id}_k \neq 0$. This is a contradiction, so the sequence does not split. $\square$

4. Find all indecomposable representations up to isomorphism of the quiver $1 \rightarrow 2$.

Proof. The indecomposable representations of $Q: 1 \rightarrow 2$ are, up to isomorphism:

$$S(1): k \xrightarrow{0} 0, \quad S(2): 0 \xrightarrow{0} k, \quad P(1): k \xrightarrow{\text{id}} k.$$

Step 1: Indecomposability.

$S(1)$ and $S(2)$ have dimension vectors $(1, 0)$ and $(0, 1)$ respectively. Since no component can be split further, these are clearly indecomposable.

For $P(1)$ with dimension vector $(1, 1)$, suppose

$$P(1) = N_1 \oplus N_2$$

is a nontrivial decomposition. Then we must have

$$\underline{\dim} N_1 = (1, 0), \quad \underline{\dim} N_2 = (0, 1)$$

(or vice versa). In this decomposition, the map $\phi: P(1)(1) = k \rightarrow P(1)(2) = k$ would decompose as

$$\phi = \phi|_{N_1} \oplus \phi|_{N_2} = 0 \oplus 0 = 0,$$

since $\phi|_{N_1}: k \rightarrow 0$ and $\phi|_{N_2}: 0 \rightarrow k$ are both the zero map. This contradicts $\phi = \text{id}_k \neq 0$. Hence $P(1)$ is indecomposable.

Step 2: Completeness.

Let

$$M = (V_1, V_2, \phi)$$

be any representation with $\phi: V_1 \rightarrow V_2$. Set

$$n_1 = \dim V_1, \quad n_2 = \dim V_2, \quad r = \text{rank}(\phi).$$

Choose a basis $\{e_1, \dots, e_{n_1}\}$ of V_1 such that $\{e_1, \dots, e_r\}$ maps to linearly independent vectors in V_2 and $\{e_{r+1}, \dots, e_{n_1}\}$ spans $\ker \phi$. Let

$$f_i = \phi(e_i) \quad \text{for } 1 \leq i \leq r,$$

and extend $\{f_1, \dots, f_r\}$ to a basis $\{f_1, \dots, f_{n_2}\}$ of V_2 . In these bases,

$$\phi = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}.$$

Then M decomposes as:

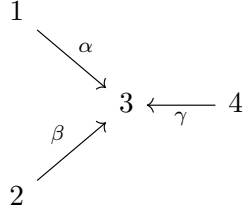
- r copies of $P(1)$: the i -th copy is spanned by e_i at vertex 1 and f_i at vertex 2, with $\phi(e_i) = f_i$, for $1 \leq i \leq r$.
- $(n_1 - r)$ copies of $S(1)$: spanned by e_{r+j} at vertex 1, with $\phi(e_{r+j}) = 0$.
- $(n_2 - r)$ copies of $S(2)$: spanned by f_{r+l} at vertex 2.

Hence

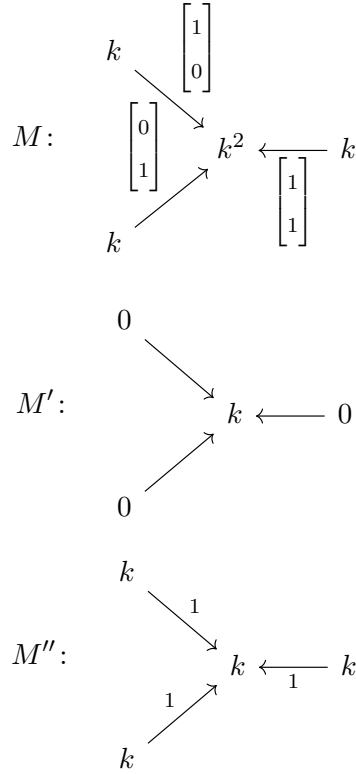
$$M \cong P(1)^{\oplus r} \oplus S(1)^{\oplus (n_1 - r)} \oplus S(2)^{\oplus (n_2 - r)}.$$

Since every representation decomposes into copies of $S(1)$, $S(2)$, and $P(1)$, these are all the indecomposable representations up to isomorphism. $\square$

5. Let Q be the quiver



and consider the following representations



(a) Show that

$$\begin{aligned}
 \operatorname{Hom}(M, M') &= 0, & \operatorname{Hom}(M', M) &\cong k^2 \\
 \operatorname{Hom}(M, M'') &\cong k^2, & \operatorname{Hom}(M'', M) &= 0.
 \end{aligned}$$

Proof. The quiver Q has arrows $\alpha: 1 \rightarrow 3$, $\beta: 2 \rightarrow 3$, $\gamma: 4 \rightarrow 3$. A morphism $f: A \rightarrow B$ consists of linear maps (f_1, f_2, f_3, f_4) satisfying

$$f_3 \circ A(\delta) = B(\delta) \circ f_s$$

for each arrow $\delta: s \rightarrow 3$.

$\operatorname{Hom}(M, M') = 0$.

A morphism $f: M \rightarrow M'$ has components

$$f_1: k \rightarrow 0, \quad f_2: k \rightarrow 0, \quad f_3: k^2 \rightarrow k, \quad f_4: k \rightarrow 0.$$

Since the codomain is 0, we have $f_1 = f_2 = f_4 = 0$. The commutativity condition at α gives

$$\begin{array}{ccc} M(1) = k & \xrightarrow{\begin{pmatrix} 1 \\ 0 \end{pmatrix}} & M(3) = k^2 \\ f_1=0 \downarrow & & \downarrow f_3 \\ M'(1) = 0 & \xrightarrow{0} & M'(3) = k \end{array} \implies f_3 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 0.$$

Similarly, the commutativity condition at β gives

$$f_3 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 0.$$

Since $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ form a basis of k^2 , we conclude $f_3 = 0$. Therefore $f = 0$.

$\text{Hom}(M', M) \cong k^2$.

A morphism $f: M' \rightarrow M$ has components

$$f_1: 0 \rightarrow k, \quad f_2: 0 \rightarrow k, \quad f_3: k \rightarrow k^2, \quad f_4: 0 \rightarrow k.$$

Since the domains of f_1, f_2, f_4 are 0, these maps are all zero. For each arrow $\delta: s \rightarrow 3$, the commutativity condition becomes

$$f_3 \circ M'(\delta) = M(\delta) \circ f_s.$$

Since $M'(s) = 0$ for $s = 1, 2, 4$, both $M'(\delta)$ and f_s are the zero map, so the condition is automatically satisfied. Hence f is completely determined by the arbitrary linear map

$$f_3 \in \text{Hom}_k(k, k^2) \cong k^2.$$

$\text{Hom}(M, M'') \cong k^2$.

A morphism $f: M \rightarrow M''$ has components

$$f_1 = a: k \rightarrow k, \quad f_2 = b: k \rightarrow k, \quad f_3 = \begin{pmatrix} c & d \end{pmatrix}: k^2 \rightarrow k, \quad f_4 = e: k \rightarrow k,$$

where $a, b, c, d, e \in k$. The commutativity conditions give:

At $\alpha: 1 \rightarrow 3$:

$$f_3 \circ M(\alpha) = M''(\alpha) \circ f_1 \implies \begin{pmatrix} c & d \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1 \cdot a \implies c = a.$$

At $\beta: 2 \rightarrow 3$:

$$f_3 \circ M(\beta) = M''(\beta) \circ f_2 \implies \begin{pmatrix} c & d \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 1 \cdot b \implies d = b.$$

At $\gamma: 4 \rightarrow 3$:

$$f_3 \circ M(\gamma) = M''(\gamma) \circ f_4 \implies \begin{pmatrix} c & d \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 1 \cdot e \implies e = a + b.$$

Hence every morphism f is determined by the pair $(a, b) \in k^2$, with $c = a$, $d = b$, $e = a + b$. Therefore

$$\text{Hom}(M, M'') \cong k^2.$$

$\text{Hom}(M'', M) = 0$.

A morphism $f: M'' \rightarrow M$ has components

$$f_1 = a: k \rightarrow k, \quad f_2 = b: k \rightarrow k, \quad f_3 = \begin{pmatrix} c \\ d \end{pmatrix}: k \rightarrow k^2, \quad f_4 = e: k \rightarrow k.$$

At $\alpha: 1 \rightarrow 3$:

$$M(\alpha) \circ f_1 = f_3 \circ M''(\alpha) \implies \begin{pmatrix} 1 \\ 0 \end{pmatrix} a = \begin{pmatrix} c \\ d \end{pmatrix} \cdot 1 \implies c = a, d = 0.$$

At $\beta: 2 \rightarrow 3$:

$$M(\beta) \circ f_2 = f_3 \circ M''(\beta) \implies \begin{pmatrix} 0 \\ 1 \end{pmatrix} b = \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} a \\ 0 \end{pmatrix} \implies a = 0, b = 0.$$

At $\gamma: 4 \rightarrow 3$:

$$M(\gamma) \circ f_4 = f_3 \circ M''(\gamma) \implies \begin{pmatrix} 1 \\ 1 \end{pmatrix} e = \begin{pmatrix} a \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \implies e = 0.$$

All components vanish, so $\text{Hom}(M'', M) = 0$. □

(b) Show that M is not isomorphic to $M' \oplus M''$.

Proof. Suppose for contradiction that

$$M \cong M' \oplus M''.$$

Then by the additivity of Hom ,

$$\text{Hom}(M, M') \cong \text{Hom}(M' \oplus M'', M') \cong \text{Hom}(M', M') \oplus \text{Hom}(M'', M').$$

Since $M' \neq 0$, the identity morphism $\text{id}_{M'}$ is a nonzero element of $\text{Hom}(M', M')$. Hence

$$\text{Hom}(M', M') \neq 0,$$

which implies

$$\text{Hom}(M, M') \neq 0.$$

This contradicts part (a), where we showed $\text{Hom}(M, M') = 0$. Therefore

$$M \not\cong M' \oplus M''.$$

□

(c) Show that there is a short exact sequence:

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0.$$

Proof. We construct explicit morphisms $\iota: M' \rightarrow M$ and $p: M \rightarrow M''$ such that

$$0 \longrightarrow M' \xrightarrow{\iota} M \xrightarrow{p} M'' \longrightarrow 0$$

is exact.

Definition of $\iota: M' \rightarrow M$. Since $M'(1) = M'(2) = M'(4) = 0$, we set

$$\iota_1 = 0: 0 \rightarrow k, \quad \iota_2 = 0: 0 \rightarrow k, \quad \iota_4 = 0: 0 \rightarrow k,$$

and define

$$\iota_3 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}: k \rightarrow k^2.$$

Since all maps of M' are zero, for each arrow $\delta: s \rightarrow 3$, the commutativity condition

$$\iota_3 \circ M'(\delta) = M(\delta) \circ \iota_s$$

becomes $0 = 0$, which is automatic. Moreover, ι_3 is injective (since $\begin{pmatrix} 1 \\ -1 \end{pmatrix} \neq 0$), so ι is injective.

Definition of $p: M \rightarrow M''$. Define

$$p_1 = 1: k \rightarrow k, \quad p_2 = 1: k \rightarrow k, \quad p_3 = \begin{pmatrix} 1 & 1 \end{pmatrix}: k^2 \rightarrow k, \quad p_4 = 2: k \rightarrow k.$$

We verify commutativity at each arrow.

At $\alpha: 1 \rightarrow 3$:

$$p_3 \circ M(\alpha) = \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1 = 1 \cdot 1 = M''(\alpha) \circ p_1.$$

At $\beta: 2 \rightarrow 3$:

$$p_3 \circ M(\beta) = \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 1 = 1 \cdot 1 = M''(\beta) \circ p_2.$$

At $\gamma: 4 \rightarrow 3$:

$$p_3 \circ M(\gamma) = \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 2 = 1 \cdot 2 = M''(\gamma) \circ p_4.$$

Each p_i is a nonzero scalar, so p is surjective at every vertex.

Exactness at M .

First, $p \circ \iota = 0$: at vertex 3,

$$p_3 \circ \iota_3 = \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 1 + (-1) = 0.$$

At other vertices, $\iota_i = 0$, so $(p \circ \iota)_i = p_i \circ 0 = 0$.

Next, we verify $\ker p = \text{im } \iota$ at each vertex:

- Vertices 1, 2, 4: each p_i is a nonzero scalar (hence injective), so $\ker p_i = 0$. Also $\iota_i = 0$, so $\text{im } \iota_i = 0$.
- Vertex 3:

$$\ker p_3 = \{(x, y) \in k^2 : x + y = 0\} = \left\langle \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\rangle = \text{im } \iota_3.$$

Therefore

$$0 \longrightarrow M' \xrightarrow{\iota} M \xrightarrow{p} M'' \longrightarrow 0$$

is a short exact sequence. □

6. Let Q be the quiver

$$1 \xrightleftharpoons[\beta]{\alpha} 2$$

For any $\lambda \in k \cup \{\infty\}$, define M_λ to be the representation:

$$M_\lambda: \quad k \xrightleftharpoons[1]{\lambda} k \quad \text{if } \lambda \in k; \quad M_\infty: \quad k \xrightleftharpoons[0]{1} k \quad \text{if } \lambda = \infty.$$

(a) Show that each M_λ is indecomposable.

Proof. Each M_λ has dimension vector

$$\underline{\dim} M_\lambda = (1, 1).$$

Suppose for contradiction that

$$M_\lambda = N_1 \oplus N_2$$

is a nontrivial decomposition (both $N_1, N_2 \neq 0$). Since $\underline{\dim} N_1 + \underline{\dim} N_2 = (1, 1)$, the only possibility is

$$\underline{\dim} N_1 = (1, 0), \quad \underline{\dim} N_2 = (0, 1)$$

(or vice versa). The summand N_2 with $N_2(1) = 0$ and $N_2(2) = k$ must be a subrepresentation. Since both arrows go from vertex 2 to vertex 1, this requires

$$M_\lambda(\alpha)(k) \subseteq N_2(1) = 0 \quad \text{and} \quad M_\lambda(\beta)(k) \subseteq N_2(1) = 0.$$

For $\lambda \in k$: the map $M_\lambda(\beta) = 1: k \rightarrow k$ is nonzero, so

$$M_\lambda(\beta)(k) = k \neq 0.$$

This is a contradiction.

For $\lambda = \infty$: the map $M_\infty(\alpha) = 1: k \rightarrow k$ is nonzero, so

$$M_\infty(\alpha)(k) = k \neq 0.$$

This is again a contradiction.

In both cases, no such subrepresentation N_2 exists, so M_λ is indecomposable. $\square$

- (b) Show that $M_\lambda \cong M_\mu$ if and only if $\lambda = \mu$. In particular, the number of indecomposable representations depends on the choice of the field k .

Proof. ($\Leftarrow$) If $\lambda = \mu$, the identity maps $f_1 = f_2 = \text{id}_k$ give an isomorphism.

($\Rightarrow$) Let $f = (f_1, f_2): M_\lambda \rightarrow M_\mu$ be an isomorphism, with

$$f_1 = a \in k^*, \quad f_2 = b \in k^*.$$

The commutativity conditions for the two arrows $\alpha, \beta: 2 \rightarrow 1$ are

$$f_1 \circ M_\lambda(\alpha) = M_\mu(\alpha) \circ f_2, \quad f_1 \circ M_\lambda(\beta) = M_\mu(\beta) \circ f_2.$$

Case 1: $\lambda, \mu \in k$. Then $M_\lambda(\alpha) = \lambda$, $M_\lambda(\beta) = 1$, $M_\mu(\alpha) = \mu$, $M_\mu(\beta) = 1$. The conditions become

$$a\lambda = \mu b, \quad a \cdot 1 = 1 \cdot b.$$

From the second equation, $a = b$. Substituting into the first:

$$a\lambda = \mu a.$$

Since $a \neq 0$, we can divide both sides by a to get $\lambda = \mu$.

Case 2: $\lambda \in k$, $\mu = \infty$. Then $M_\lambda(\beta) = 1$ and $M_\infty(\beta) = 0$. The commutativity at β gives

$$a \cdot 1 = 0 \cdot b = 0,$$

so $a = 0$, contradicting $a \in k^*$. Hence no isomorphism exists.

The case $\lambda = \infty$, $\mu \in k$ is symmetric.

Case 3: $\lambda = \mu = \infty$. Trivially $\lambda = \mu$.

In all cases, $M_\lambda \cong M_\mu$ implies $\lambda = \mu$. Since

$$|k \cup \{\infty\}| = |k| + 1,$$

the number of pairwise non-isomorphic indecomposable representations of this form depends on the cardinality of the field k . $\square$

(c) Show that $\text{Hom}(M_\lambda, M_\mu) = 0$ if $\lambda \neq \mu$.

Proof. Let $f = (f_1, f_2): M_\lambda \rightarrow M_\mu$ be a morphism, with $f_1 = a$ and $f_2 = b$ (scalars). The commutativity conditions are

$$f_1 \circ M_\lambda(\alpha) = M_\mu(\alpha) \circ f_2, \quad f_1 \circ M_\lambda(\beta) = M_\mu(\beta) \circ f_2.$$

Case 1: $\lambda, \mu \in k$. The conditions become

$$a\lambda = \mu b, \quad a = b.$$

Substituting $b = a$ into the first equation:

$$a\lambda = \mu a \implies a(\lambda - \mu) = 0.$$

Since $\lambda \neq \mu$, we have $\lambda - \mu \neq 0$, so $a = 0$. Then $b = a = 0$.

Case 2: $\lambda \in k, \mu = \infty$. Then $M_\lambda(\beta) = 1, M_\infty(\beta) = 0$. The condition at β gives

$$a \cdot 1 = 0 \cdot b = 0,$$

so $a = 0$. The condition at α then gives

$$0 \cdot \lambda = 1 \cdot b \implies b = 0.$$

Case 3: $\lambda = \infty, \mu \in k$. Then $M_\infty(\beta) = 0, M_\mu(\beta) = 1$. The condition at β gives

$$a \cdot 0 = 1 \cdot b \implies b = 0.$$

The condition at α gives

$$a \cdot 1 = \mu \cdot 0 = 0 \implies a = 0.$$

In every case, $f_1 = f_2 = 0$, so $f = 0$. Therefore $\text{Hom}(M_\lambda, M_\mu) = 0$. $\square$

(d) Show that for each λ there is a short exact sequence

$$0 \longrightarrow S(1) \longrightarrow M_\lambda \longrightarrow S(2) \longrightarrow 0,$$

where $S(1)$ and $S(2)$ are the representations

$$S(1): \quad k \xleftarrow{\quad} 0 \qquad S(2): \quad 0 \xleftarrow{\quad} k$$

Proof. Define $\iota: S(1) \rightarrow M_\lambda$ and $p: M_\lambda \rightarrow S(2)$ by

$$\begin{aligned}\iota_1 &= \text{id}_k: k \rightarrow k, & \iota_2 &= 0: 0 \rightarrow k, \\ p_1 &= 0: k \rightarrow 0, & p_2 &= \text{id}_k: k \rightarrow k.\end{aligned}$$

Morphism verification. Both arrows α, β go from vertex 2 to vertex 1. For ι , the commutativity diagram at each arrow $\delta \in \{\alpha, \beta\}$ is

$$\begin{array}{ccc} S(1)(2) = 0 & \xrightarrow{0} & S(1)(1) = k \\ \iota_2=0 \downarrow & & \downarrow \iota_1=\text{id}_k \\ M_\lambda(2) = k & \xrightarrow{M_\lambda(\delta)} & M_\lambda(1) = k \end{array}$$

Both compositions give 0, so the diagram commutes.

For p , the commutativity diagram at each arrow δ is

$$\begin{array}{ccc} M_\lambda(2) = k & \xrightarrow{M_\lambda(\delta)} & M_\lambda(1) = k \\ p_2=\text{id}_k \downarrow & & \downarrow p_1=0 \\ S(2)(2) = k & \xrightarrow{0} & S(2)(1) = 0 \end{array}$$

Both compositions give 0, so the diagram commutes.

Exactness.

ι is injective: $\iota_1 = \text{id}_k$ is injective.

p is surjective: $p_2 = \text{id}_k$ is surjective.

Exactness at M_λ : we verify $\ker p = \text{im } \iota$ at each vertex.

- Vertex 1:

$$\ker(p_1: k \rightarrow 0) = k = \text{im}(\iota_1) = \text{im}(\text{id}_k) = k.$$

- Vertex 2:

$$\ker(p_2) = \ker(\text{id}_k) = 0 = \text{im}(\iota_2) = \text{im}(0) = 0.$$

Hence

$$0 \longrightarrow S(1) \xrightarrow{\iota} M_\lambda \xrightarrow{p} S(2) \longrightarrow 0$$

is a short exact sequence. □

- 7.** Let $f: M \rightarrow N$ be a morphism of representations. Prove that the definition of $\text{coker } f$ is equivalent to the one defined in classroom through universal property (otherwise, see Remark 1.6 in Schiffler's book).

Proof. Let

$$f = (f_i): M = (M_i, \varphi_\alpha) \rightarrow N = (N_i, \psi_\alpha)$$

be a morphism of representations. For each vertex i , define

$$H_i := N_i / f_i(M_i).$$

For each arrow $\alpha : i \rightarrow j$, define

$$\chi_\alpha(n_i + f_i(M_i)) := \psi_\alpha(n_i) + f_j(M_j).$$

Hence

$$H = (H_i, \chi_\alpha)$$

is a representation.

Now let

$$g = (g_i) : N \rightarrow H$$

be the natural quotient morphism, where

$$g_i : N_i \rightarrow N_i/f_i(M_i), \quad g_i(n_i) = n_i + f_i(M_i).$$

Then for every vertex i and every $m_i \in M_i$,

$$(g_i \circ f_i)(m_i) = f_i(m_i) + f_i(M_i) = 0.$$

Thus

$$g \circ f = 0.$$

We now prove the universal property. Let

$$h = (h_i) : N \rightarrow Y = (Y_i, \eta_\alpha)$$

be a morphism such that

$$h \circ f = 0.$$

Then for each vertex i ,

$$h_i \circ f_i = 0.$$

Hence h_i vanishes on $f_i(M_i)$, so it factors uniquely through the quotient $N_i/f_i(M_i)$.

Therefore there exists a unique linear map

$$u_i : H_i = N_i/f_i(M_i) \rightarrow Y_i$$

such that

$$u_i(n_i + f_i(M_i)) := h_i(n_i).$$

This is well-defined because if

$$n_i + f_i(M_i) = n'_i + f_i(M_i),$$

then $n_i - n'_i = f_i(m_i)$ for some $m_i \in M_i$, and hence

$$h_i(n_i) - h_i(n'_i) = h_i(f_i(m_i)) = 0.$$

By construction,

$$u_i \circ g_i = h_i$$

for every vertex i .

It remains to check that $u = (u_i) : H \rightarrow Y$ is a morphism of representations. Let $\alpha : i \rightarrow j$ be an arrow and let $n_i + f_i(M_i) \in H_i$. Then

$$\begin{aligned} \eta_\alpha(u_i(n_i + f_i(M_i))) &= \eta_\alpha(h_i(n_i)) = h_j(\psi_\alpha(n_i)) = u_j(\psi_\alpha(n_i) + f_j(M_j)) \\ &= u_j(\chi_\alpha(n_i + f_i(M_i))). \end{aligned}$$

Therefore

$$\eta_\alpha \circ u_i = u_j \circ \chi_\alpha,$$

so u is a morphism of representations.

Finally, uniqueness is clear: if

$$u' = (u'_i) : H \rightarrow Y$$

also satisfies

$$u' \circ g = h,$$

then for each i ,

$$u'_i(n_i + f_i(M_i)) = u'_i(g_i(n_i)) = h_i(n_i) = u_i(n_i + f_i(M_i)).$$

Thus $u'_i = u_i$ for all i , so $u' = u$.

Therefore the quotient representation

$$(N_i/f_i(M_i), \chi_\alpha)$$

with the natural projection $g : N \rightarrow H$ satisfies the universal property of the cokernel.

On the other hand, suppose

$$g = (g_i) : N = (N_i, \psi_\alpha) \rightarrow H = (H_i, \chi_\alpha)$$

is a cokernel of

$$f = (f_i) : M = (M_i, \varphi_\alpha) \rightarrow N = (N_i, \psi_\alpha)$$

in the sense of the universal property.

Let

$$C := (N_i/f_i(M_i), \bar{\psi}_\alpha)$$

be the pointwise cokernel representation, where

$$\bar{\psi}_\alpha(n_i + f_i(M_i)) := \psi_\alpha(n_i) + f_j(M_j).$$

Let

$$\pi = (\pi_i) : N \rightarrow C$$

be the natural quotient morphism. As shown above,

$$\pi \circ f = 0.$$

Since $g : N \rightarrow H$ is a cokernel of f , applying its universal property to the morphism

$$\pi : N \rightarrow C$$

with $\pi \circ f = 0$, there exists a unique morphism

$$a : H \rightarrow C$$

such that

$$a \circ g = \pi.$$

On the other hand, since $g \circ f = 0$, for each vertex i we have

$$g_i \circ f_i = 0.$$

Thus g_i vanishes on $f_i(M_i)$, so it factors uniquely through the quotient

$$N_i/f_i(M_i).$$

Hence for each i there exists a unique linear map

$$b_i : C_i = N_i/f_i(M_i) \rightarrow H_i$$

given by

$$b_i(n_i + f_i(M_i)) := g_i(n_i).$$

These maps assemble into a morphism

$$b : C \rightarrow H,$$

because for any arrow $\alpha : i \rightarrow j$,

$$\begin{aligned} \chi_\alpha(b_i(n_i + f_i(M_i))) &= \chi_\alpha(g_i(n_i)) \\ &= g_j(\psi_\alpha(n_i)) \\ &= b_j(\psi_\alpha(n_i) + f_j(M_j)) \\ &= b_j(\overline{\psi}_\alpha(n_i + f_i(M_i))). \end{aligned}$$

So

$$b \circ \pi = g.$$

We now show that a and b are inverse isomorphisms.

First,

$$(a \circ b) \circ \pi = a \circ (b \circ \pi) = a \circ g = \pi.$$

Also,

$$\text{id}_C \circ \pi = \pi.$$

Since $\pi : N \rightarrow C$ is a cokernel of f , the uniqueness part of its universal property implies

$$a \circ b = \text{id}_C.$$

Similarly,

$$(b \circ a) \circ g = b \circ (a \circ g) = b \circ \pi = g,$$

and also

$$\text{id}_H \circ g = g.$$

Since $g : N \rightarrow H$ is a cokernel of f , the uniqueness part of its universal property implies

$$b \circ a = \text{id}_H.$$

Therefore a and b are inverse isomorphisms, and hence

$$H \cong C = (N_i/f_i(M_i), \overline{\psi}_\alpha).$$

In particular, for each vertex i we have

$$H_i \cong N_i/f_i(M_i),$$

and under these identifications, for every arrow $\alpha : i \rightarrow j$,

$$\chi_\alpha = a_j^{-1} \circ \overline{\psi}_\alpha \circ a_i.$$

Thus any cokernel given by the universal property is isomorphic to the pointwise quotient representation

$$(N_i/f_i(M_i), \psi_\alpha \text{ induced on the quotients}).$$

□

8. Let M, M' and N be representations of Q and let $f : M \rightarrow N, g : M' \rightarrow N$ be morphisms. Define the *fiber product* (or *pull back*) of f and g as

$$X = \{(a, b) \mid a \in M, b \in M', \text{ such that } f(a) = g(b)\},$$

and define the projections $\pi_1(a, b) = a$ and $\pi_2(a, b) = b$.

- (a) Show that X is a subrepresentation of $M \oplus M'$ and that there is a commutative

diagram:

$$\begin{array}{ccc} X & \xrightarrow{\pi_2} & M' \\ \pi_1 \downarrow & & \downarrow g \\ M & \xrightarrow{f} & N \end{array}$$

Proof. **X is a subrepresentation of $M \oplus M'$.**

At each vertex i , define

$$X_i = \{(a, b) \in M_i \oplus M'_i : f_i(a) = g_i(b)\}.$$

This is the kernel of the linear map

$$M_i \oplus M'_i \longrightarrow N_i, \quad (a, b) \longmapsto f_i(a) - g_i(b),$$

so X_i is a linear subspace of $M_i \oplus M'_i$.

We now show that for any arrow $\alpha: i \rightarrow j$, the map $(M \oplus M')(\alpha)$ sends X_i into X_j . Let $(a, b) \in X_i$, so

$$f_i(a) = g_i(b).$$

We need to show $(M(\alpha)(a), M'(\alpha)(b)) \in X_j$, i.e., $f_j(M(\alpha)(a)) = g_j(M'(\alpha)(b))$.

Using the commutativity of f and g :

$$f_j(M(\alpha)(a)) = N(\alpha)(f_i(a)) = N(\alpha)(g_i(b)) = g_j(M'(\alpha)(b)).$$

Hence X is stable under the maps of $M \oplus M'$ and is therefore a subrepresentation.

Commutativity of the diagram.

For any $(a, b) \in X_i$, we have

$$(f \circ \pi_1)_i(a, b) = f_i(\pi_1(a, b)) = f_i(a) = g_i(b) = g_i(\pi_2(a, b)) = (g \circ \pi_2)_i(a, b).$$

Hence $f \circ \pi_1 = g \circ \pi_2$, and the diagram

$$\begin{array}{ccc} X & \xrightarrow{\pi_2} & M' \\ \pi_1 \downarrow & & \downarrow g \\ M & \xrightarrow{f} & N \end{array}$$

commutes. □

(b) Show that if f is injective then π_2 is injective.

Proof. Suppose f is injective, i.e., f_i is injective for each vertex i .

Let $(a, b) \in X_i$ with

$$\pi_2(a, b) = b = 0.$$

Since $(a, b) \in X_i$, we have $f_i(a) = g_i(b)$. Substituting $b = 0$:

$$f_i(a) = g_i(0) = 0.$$

Since f_i is injective, it follows that

$$a = 0.$$

Hence $(a, b) = (0, 0)$, so $\ker(\pi_2)_i = 0$ at every vertex i . Therefore π_2 is injective. $\square$

(c) Show that if f is surjective then π_2 is surjective.

Proof. Suppose f is surjective, i.e., f_i is surjective for each vertex i .

Let $b \in M'_i$. We need to find $a \in M_i$ such that $(a, b) \in X_i$, i.e., such that

$$f_i(a) = g_i(b).$$

The element $g_i(b) \in N_i$. Since f_i is surjective, we have

$$N_i = \text{im}(f_i),$$

so there exists $a \in M_i$ with $f_i(a) = g_i(b)$. Then $(a, b) \in X_i$ by definition, and

$$\pi_2(a, b) = b.$$

Since $b \in M'_i$ was arbitrary, π_2 is surjective at each vertex. $\square$

(d) Suppose $0 \rightarrow L \xrightarrow{h} M \xrightarrow{f} N \rightarrow 0$ is a short exact sequence, define $h': L \rightarrow X$ by $h'(n) = (h(n), 0)$. Show that the following diagram is commutative with exact rows:

$$\begin{array}{ccccccc} 0 & \longrightarrow & L & \xrightarrow{h'} & X & \xrightarrow{\pi_2} & M' \longrightarrow 0 \\ & & \downarrow 1_L & & \downarrow \pi_1 & & \downarrow g \\ 0 & \longrightarrow & L & \xrightarrow{h} & M & \xrightarrow{f} & N \longrightarrow 0 \end{array}$$

Proof. **h' is well-defined.**

For $n \in L_i$, we must check that $(h_i(n), 0) \in X_i$, i.e., that

$$f_i(h_i(n)) = g_i(0).$$

By exactness of the bottom row at M , we have $f \circ h = 0$, so

$$f_i(h_i(n)) = 0 = g_i(0).$$

Hence $h'_i(n) = (h_i(n), 0) \in X_i$.

h' is a morphism of representations.

For each arrow $\alpha: i \rightarrow j$ and $n \in L_i$:

$$\begin{aligned} X(\alpha)(h'_i(n)) &= X(\alpha)(h_i(n), 0) \\ &= (M(\alpha)(h_i(n)), M'(\alpha)(0)) \\ &= (h_j(L(\alpha)(n)), 0) \\ &= h'_j(L(\alpha)(n)), \end{aligned}$$

where the third equality uses the commutativity of h , i.e., $M(\alpha) \circ h_i = h_j \circ L(\alpha)$.

Commutativity of the diagram.

Left square: for each $n \in L_i$,

$$(\pi_1 \circ h')_i(n) = \pi_1(h_i(n), 0) = h_i(n) = (h \circ 1_L)_i(n).$$

So $\pi_1 \circ h' = h \circ 1_L$.

Right square: $f \circ \pi_1 = g \circ \pi_2$ was established in part (a).

Exactness of the top row.

h' is injective: suppose

$$h'_i(n) = (h_i(n), 0) = (0, 0).$$

Then $h_i(n) = 0$. Since h is injective (from the exactness of the bottom row),

$$n = 0.$$

Hence h' is injective.

π_2 is surjective: since f is surjective (from the bottom row), this follows from part (c).

$\text{im}(h') = \ker(\pi_2)$:

($\supseteq$) For any $n \in L_i$,

$$\pi_2(h'_i(n)) = \pi_2(h_i(n), 0) = 0,$$

so $\text{im}(h'_i) \subseteq \ker(\pi_2)_i$.

($\subseteq$) Let $(a, b) \in X_i$ with $\pi_2(a, b) = b = 0$. Then

$$f_i(a) = g_i(0) = 0,$$

so $a \in \ker f_i$. By exactness of the bottom row at M ,

$$\ker f_i = \text{im } h_i.$$

Hence there exists $n \in L_i$ with $a = h_i(n)$, and then

$$(a, 0) = (h_i(n), 0) = h'_i(n) \in \text{im}(h'_i).$$

Therefore $\text{im}(h') = \ker(\pi_2)$, and the top row is exact. □

9. Let $f: L \rightarrow M$ be a morphism. Then a *cokernel* of f is a morphism $g: M \rightarrow N$ such that $gf = 0$, and given any morphism $v: M \rightarrow X$ such that $vf = 0$, there is a unique morphism $u: N \rightarrow X$ such that $ug = v$:

$$\begin{array}{ccccc} L & \xrightarrow{f} & M & \xrightarrow{g} & N \\ & & \downarrow v & \swarrow \exists u & \\ & & X & & \end{array}$$

Prove that the two definitions of cokernel agree.

10. Let Q be a quiver and let

$$L \xrightarrow{f} M \xrightarrow{g} N \longrightarrow 0$$

be a sequence in $\text{Rep } Q$. Show that this sequence is exact if and only if for every representation $X \in \text{Rep } Q$, the sequence

$$0 \longrightarrow \text{Hom}(N, X) \xrightarrow{g^*} \text{Hom}(M, X) \xrightarrow{f^*} \text{Hom}(L, X)$$

is exact, where $g^*(h) = h \circ g$ and $f^*(k) = k \circ f$.

11. Let $g: M \rightarrow N$ be a surjective morphism between representations of Q , and let $P(i)$ be the projective representation at vertex i . Show that the map

$$g_*: \text{Hom}(P(i), M) \rightarrow \text{Hom}(P(i), N), \quad \varphi \longmapsto g \circ \varphi$$

is surjective.

12. Let $g: L \rightarrow M$ be an injective morphism between representations of Q , and let $I(i)$ be the injective representation at vertex i . Show that the map

$$g^*: \text{Hom}(M, I(i)) \rightarrow \text{Hom}(L, I(i)), \quad \psi \longmapsto \psi \circ g$$

is surjective.

13. Let Q be the quiver

$$\begin{array}{ccccc} 1 & \xrightarrow{\quad} & 3 & \longrightarrow & 4 \\ & \searrow & \nearrow & & \\ & 2 & & & \end{array}$$

Prove that

$$\begin{array}{ccccc} & & \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} & & \\ & & \downarrow & & \\ P(1) \simeq k & \xrightarrow{\quad} & k^2 & \xrightarrow{\begin{bmatrix} a & b \\ c & d \end{bmatrix}} & k^2 \\ & \searrow z & \nearrow \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} & & \\ & & k & & \end{array}$$

if and only if all four maps have maximal rank, that is,

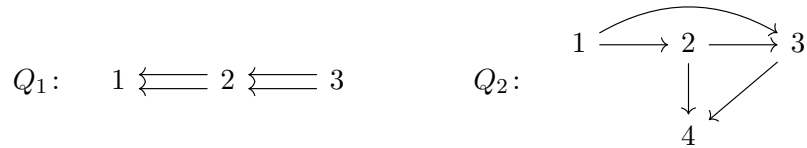
$$ad - bc \neq 0, \quad (x_1, x_2) \neq (0, 0), \quad (y_1, y_2) \neq (0, 0), \quad z \neq 0$$

and the vectors

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

are linearly independent.

- 14.** Compute the indecomposable projective (resp. injective) representations $P(i)$ (resp. $I(i)$) for the following quivers:



- 15.** Compute a projective resolution of each simple representation $S(i)$ for the quivers Q_1 and Q_2 from Problem 14.